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Affinity Propagation Clustering of Full-field, High-spatial-dimensional Measurements for Robust Output-only Modal Identification: A Proof-of-Concept Study

Yongchao Yang[1,*], Charles Dorn[2]

Abstract

Determination of the model order is a challenging problem in system identification, especially in output-only or operational modal identification where some modes are weakly excited. Although existing methods such as the stabilization diagram method (spectral information) are effective, they do not scale to high-dimensional data, which is usually needed for high-fidelity characterization of structural dynamics and has been made available in the emerging full-field measurement techniques using optical methods such as photogrammetry and laser vibrometers.

In this proof-of-concept study we present a new non-parametric, data-driven approach for *robust* output-only identification of high-spatial-dimensional modal parameters of basic structures by efficiently processing and interactively exploiting the full-field measurement (i.e., very dense spatial measurement points). Specifically, we first *over-estimate* the system model *once*, producing a pool of candidate modes associated with their modal frequencies and full-field, high-spatial-dimensional mode shapes. This is accomplished by a data-driven method termed affinity propagation clustering (APC), where the active clusters, which are the active

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modes in our formulations, emerge from the "message-passing" procedure and does not require a pre-determination of the cluster number (mode or model order). Next, rather than using the *spectral* information to distinguish the physical and spurious modes in the stabilization diagram method, we exploit and visualize the *spatial*, full-field mode shape associated with each candidate mode to do so. We conduct extensive experiments on basic structural models with comparisons to a few existing methods. The results indicate that the new method is computationally efficient for identifying high-spatial-dimensional modal parameters, and robust to identify weak modes by exploiting the full-field measurement. We also discuss its applicability and limitations for structures with complex geometry (shapes).

Keywords: Structural Dynamics • Full-field Measurement • Operational Modal Analysis • Clustering • Visualization • Video Processing

1. Introduction

Recent research and practice have significantly advanced full-field, high-spatial-density vibration measurements for structural dynamics and health monitoring. Its benefits are significant: it allows high-fidelity model development, validation, and updating of structural dynamics [1] [2]. In addition, high-resolution mode shapes, for example, enable accurate localization of structural damage, which is typically a local phenomenon [3] [4]. Among others, non-contact optical measurement techniques have been widely studied to obtain full-field vibration measurements for *experimental* modal analysis, e.g., interferometry techniques such as laser Doppler vibrometers [5] and photogrammetry using white light illuminations, [6] such as digital image correlation (DIC) [7] [8] [9] and optical flow [10] [11] [12] [13].

Extracting the full-field modal parameters from the high-dimensional vibration response measurement of output-only structures is challenging. Many parameterized mathematical models

have been developed, mostly for traditional discrete-type, low-spatial-dimensional modal analysis, such as the classical autoregressive moving average (ARMA) model [14] and state-space model based methods such as Polyreference or PolyMAX [15] and stochastic subspace identification (SSI) [16]. Non-parametric methods also exist such as complex mode indicator function (CMIF) and frequency domain decomposition (FDD) [17], both of which are based on the response spectral matrix, and blind source separation (BSS) [18] [19]. Recently, a data-driven method [12] exploits the relation between the (high-dimensional) modal superposition model and the linear mixture models of a family of blind source separation models to efficiently process the high spatial (pixel) dimensional structural response for output-only modal identification. First, high-dimensional structural response is linearly projected by the principal component analysis (PCA) onto just few *significant* principal components that correspond to the mixtures of the active modes, drastically reducing the dimensionality. Then, the principal components are further linearly separated by blind source separation (BSS) into individual modes, whereby estimating the output-only structure's modal frequencies, damping ratios, and full-field (full-pixel-resolution) mode shapes (through inverse linear transforms of PCA-BSS). Such a method has also been adapted to extract high-dimensional modal parameters from neuromorphic event-based imaging [20] and cable vibration videos [21].

A common challenge, however, lies in the determination of the active mode number, especially when the structural response measurement data (output-only) is noisy. An inappropriate selection of the model order results in either mistakenly identifying spurious modes as physical modes or missing weakly-excited modes. Such a problem (the selection of the model order) challenges the parametric model based methods. The most widely accepted tool to address this problem is using a stabilization diagram. It is constructed by repeatedly performing system

identification with different, mostly over-specified model orders, and plotting the model order versus the corresponding identified modal frequencies. The basic, *empirical* assumption is that as model order varies, physical modes appear to have stable modal frequencies, which is not the case for spurious modes. However, interpretation of the stabilization diagram, i.e., distinguishing the physical and spurious modes, remains a challenging task as it requires extensive user expertise and experience. Although significant progress has been made to automate this interpretation process [22] [23] [24] [25] [26] [27] [28] [29] [30], the required step of over-specifying a variety of model orders for (repeatedly) processing high-dimensional data is computationally extremely exhaustive. For non-parametric response spectral matrix based methods, such as CMIF and FDD [17] [31], it's also a daunting task to determine the correct (complete) active modal peaks from the spectral matrix, whose computation itself is also extremely burdensome for high-dimensional data. An essentially equivalent problem exists in the data-driven PCA-BSS method, that is, determination of the significant principal components (associated with the significant, non-spurious singular values corresponding to the active modes).

In this proof-of-concept work we devise a new non-parametric, data-driven approach for *robust* output-only identification of high-dimensional modal parameters, by efficiently processing and interactively exploiting the full-field measurement. Specifically, we first *overestimate* the system model *once*, producing a pool of candidate modes associated with their modal frequencies and full-field, high-spatial-dimensional mode shapes. This is accomplished by a data-driven method termed affinity propagation clustering (APC)) [32], where the active clusters, which are the active modes in our formulations, emerge from the "message-passing" procedure and does not require a pre-determination of the cluster (mode or model order) number. Rather than using the spectral information (modal frequencies) to distinguish the physical and

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spurious modes in the stabilization diagram method, we instead visualize the spatial, full-field mode shape associated with each candidate mode to do so.

Experimental results indicate that the new method is able to identifying weak modes by exploiting the full-field measurement (i.e., very dense spatial measurement points). Furthermore, we also demonstrate that our new method is computationally efficient and advantageous for output-only identification of high-dimensional modal parameters. We conduct extensive experiments on basic structural models with comparisons with a few existing methods and discuss its performance and applicability (i.e., its limitations for structures with complex geometry or shape).

2. Approach

This section presents the development of the new method for output-only identification of high-dimensional modal parameters. The overall flowchart is shown in Fig. 1 and the details are described in the following.

Structural vibration response (displacements) $\delta(x,t)$ with a spatial dimension N (could be the degree-of-freedom number or *practically* the spatial measurement number of the degree-of-freedom) is represented by modal superposition as linear mixtures of n modal coordinates $q_i(t)$ associated with mode shapes N-dimensional $\varphi_i(x)$ of the ith mode

$$\delta(x,t) = \Phi(x)q(t) = \sum_{i=1}^{n} \varphi_i(x)q_i(t) \tag{1}$$

where $\Phi(x)$ is the mode shape matrix and q is the n-dimensional modal coordinate vector. Output-only modal identification aims to identify both the mode shapes $\varphi_i(x)$ and modal coordinates $q_i(t)$ that contain the information of modal frequencies ω_i and damping ratios ζ_i, using only the structural response $\delta(x,t)$.

Because the spatial dimension of $\delta(x,t)$, N, is much higher than the mode dimension (number), n, i.e., $N \gg n$, Eq. (1) is a high-spatial-dimensional, low-modal-dimensional *over-complete* model representation of $\delta(x,t)$. In addition, n is usually very small, i.e., there are usually just few active modes present in $\delta(x,t)$.

2.1. Space-frequency representation of high-spatial-dimensional, full-field structural response

Performing frequency transformation of structural response at each spatial dimension (sensor/pixel) yields a space-frequency representation $\delta(x,\omega)$ [3]of the structural response and Eq. (1) becomes

$$\delta(x,\omega) = \Phi(x)q(\omega) = \sum_{i=1}^{n} \varphi_i(x)q_i(\omega) \tag{2}$$

Note that modal coordinates $q(\omega)$ are monotone (for any i, $q_i(\omega) = 0$ except $q_i(\omega_i)$) and spectrally disjoint, i.e., for any $j \neq i$, $\omega_j \neq \omega_i$. Therefore, at ω_i $(i = 1, \dots, n)$, Eq. (2) further becomes

$$\delta(x,\omega) = \sum_{i=1}^{n} \varphi_i(x)q_i(\omega_i) \tag{3}$$

and elsewhere $\delta(x,\omega) = 0$. Because $q_i(\omega_i)$ is a only scalar, $\delta(x,\omega)$ then becomes n (sparse) points in the N-dimensional spatial (pixel) domain $x \coloneqq (x^1, x^2, \dots, x^N)$, and for each i, its coordinate is the mode shape: $\delta_i(x) \coloneqq (\delta_i^1, \delta_i^2, \dots, \delta_i^N) = \varphi_i(x)$. See Figure 2 for illustrations.

Remarks: Since $\delta(x,\omega)$ is always noisy, in practice, instead of n points there will be n modal clusters with noise as the neighborhood points around each of the modal cluster. Therefore, the problem of output-only modal identification has been distilled to identifying the clusters from the high-dimensional space-frequency representation of the structural response, where the frequency

[3] $\delta(x,\omega)$ carrying the frequency variable f denotes the Fourier or Cosine transform of $\delta(x,t)$ w.r.t. the time variable t. This also applies to other transformed variables in this paper.

index and the spatial coordinate of each identified cluster are the modal frequency ω_i and the high-spatial-dimensional mode shape $\varphi_i(x)$. Since N, the pixel number or spatial dimension, is very high, the cluster centers may not be identified by scatter plots and high-dimensional clustering analysis is needed. We adopt a data-driven affinity propagation clustering method [32] to accomplish this, as described in the following.

2.2. Affinity propagation clustering (APC) of high-dimensional structural data

With a set of spatial-frequency data points $\delta_k(x)$ (denoting $\delta(x,\omega)$ at the frequency index $k \coloneqq \omega_k$), a measure of pairwise similarity is defined as the negative squared distance $s(k,l) = -\|\delta_k - \delta_l\|^2$. The diagonal, $s(k,k)$, represents the input *preference* of δ_k to become an exemplar; it is usually set to an *a priori* universal default constant (i.e., a usually reliable choice is the median similarity of all pairs of data points) [32]. This indicates that each data point initially is *equally* suitable (same preference) to be an exemplar, which is especially the case for blind, output-only system identification in this study.

Different from other clustering based methods that require the number of clusters to be pre-specified, the clusters in APC are influenced by the input preference, and emerge from the "message-propagation" procedure. It updates iteratively two message matrices:

(1) the responsibility $r(k,l)$ quantifies how suitable for δ_l to serve as the exemplar for δ_k, in competition with other candidate exemplar $\delta_{l'}$ for δ_k;

(2) the availability $a(k,l)$ quantifies how appropriate for δ_k to select δ_l as its exemplar, in competition with other point $\delta_{k'}$'s preference for δ_l as an exemplar.

The updating rule is as follows (Figure 3):

$$r(k,l) \leftarrow s(k,l) - \max_{l' \neq l}\{a(k,l') + s(k,l')\} \tag{1}$$

$$a(k,l) \leftarrow \min\left(0, r(l,l) + \sum_{k' \notin \{k,l\}} \max(0, r(k',l))\right), \text{ for } k \neq l$$

$$a(l,l) \leftarrow \sum_{k' \neq l} \max(0, r(k',l))$$

The iterative update starts with $a(k,l) = 0$ for all k, l and undergoes until either the clusters remain unchanged over a number of iterations or after some predetermined number of iterations. As these update rules involve only simple local computation where messages are exchanged only between pairs of points, it is completely data-driven and can be easily and fast implemented and particularly suitable for processing the high-dimensional data in this study.

At any step during the affinity propagation, larger combined availability and responsibility indicates more evidence for point δ_l to be an exemplar. The final exemplars are then assigned to those δ_l that yields largest combined availability and responsibility $r(k,l) + a(k,l) > 0$. Thus, each APC-identified exemplar (cluster) is a *candidate* mode, with the mode shape $\delta_l(x)$ and the modal frequency f_l.

Remarks: As mentioned in Section 2.2 and will be presented in Section 3.2 (and found in our study), setting the input *preference* of δ_k, $s(k,k)$, to an *a priori* universal default constant (the median similarity of all pairs of data points, which tends to be conservative) reliably produces a pool of overestimated candidate modes. It should be mentioned that changing such a preference parameter will vary the overestimation number of the clusters (candidate modes). However, since APC always uses the existing data points as candidate clusters, changing the preference parameters will only add or remove other less significant clusters to the *existing* dominant clusters. Therefore, APC is robust in producing of a pool of overestimated candidates that include the dominant genuine modes.

2.3. Distinguishing genuine and spurious modes with identified full-field mode shapes

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With a pool of *candidate* modes identified by APC, each of which is associated with its corresponding (high-dimensional) mode shape $\delta_l(x)$ and the modal frequency ω_l, it is straightforward to distinguish the genuine physical and spurious modes by visualizing their associated full-field, high-spatial-dimensional mode shapes. Many, if not most, engineering structures (bridges, buildings, mechanical components, etc.) have well-defined, easily-recognizable mode shapes. Therefore, compared to the stabilization diagram using the modal frequency information, visualizing the full-field, high-spatial-dimensional is a user-friendly and relatively reliable process.

Limitations: For geometrically complex or irregular structures, their mode shapes are not easily recognized. One feasible solution for such structures would be building a *rough* finite element model, from which their analytical mode shapes are available for the user to correlate with the APC estimated full-field mode shapes for distinguishing the genuine and spurious modes. On the one hand, mode shapes are only dependent on the mass and stiffness distribution that are relatively easier to obtain in practice. On the other, the full-field, high-dimensional or high-fidelity mode shapes also makes such correlations more reliable.

3. Experimental validations

3.1. Laboratory experimental setup

In our first experiment, a bench-scale aluminum cantilever beam structure was used to validate the proposed approach. An impact hammer was used to excite (impact force) the beam horizontally at close to the fixed end. A stationary camera Sony NXCAM outfitted by a Zeiss lens with a fixed focal length of 24 mm performed video measurements of the vibrating beam at a frame rate of 480 frames per second (480 Hz frame rate and shutter speed is 1/750 second). The illumination environment is the ambient indoor fluorescent office lighting without any

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external illumination enhancement. For validations and comparisons, experimental results were compared to analytical cantilever beam bending mode shapes [33]. Figure 4 shows the video measurement of the experimental beam model.

3.2. Implementation, results, comparisons, and discussions

3.2.1. Estimating a pool of candidate full-field modes from the digital video of output-only structures

The video of the beam vibration consisting of 600 frames with a 384 ×216 pixel density was used. Phase time histories, which are proportional to the displacement time histories, are first extracted by the phase-based optical flow method (detailed in our recent work [12]) and transformed to the frequency domain by discrete cosine transform (DCT). Figure 5 shows the displacement time histories at 3 arbitrarily selected pixel locations, and Figure 6 shows the scatter plot (limited to 3-dimensional for illustrations) of their DCT. It is observed that 3 dominant modes stand out, while other possible weakly excited modes are non-observable.

The DCTs are then input into APC with the default median similarity as the approximate of the preference, producing a pool of 9 clusters as the estimate of the candidate modes with the corresponding modal frequencies (the location of the cluster in the frequency axis) and its associated full-field mode shape, as shown in Fig. 8. By visualizing the *full-field* mode shapes, it is quite straightforward to distinguish genuine and spurious modes: #1 (7.68 Hz), #5 (48 Hz), #8 (132.96 Hz), and #9 (219.84 Hz) clusters have well-behaved associated mode shapes, whereas the rest of the clusters have noise-like, scattering shapes. These identified modes also correlate well with the analytical modes, as seen in Table 1.

3.2.2. The effects of the preference parameter

The effects of varying the preference parameter is studied by setting a coefficient of the default median similarity. In general, decreasing the preference parameter will increase the

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overestimation of the clusters (candidate modes). It is found that while a different preference parameter produces a different set of candidate modes, the method gives a very robust estimation of the active *genuine* modes when the preference parameter is set in the neighborhood of the median similarity. Fig. A2 and Fig. A3 present 11 and 8 candidate modes for coefficients of 0.7 and 1.5, respectively. It is also clear that by visualizing the full-field mode shapes the genuine modes can be easily distinguished: #1 (7.68 Hz), #7 (48 Hz), #10 (132.96 Hz), and #11 (219.84 Hz) in the 0.7 case (Fig. A2), and #1 (7.68 Hz), #5 (48 Hz), #7 (132.96 Hz), and #8 (219.84 Hz) in the 1.5 cases (Fig. A3).

On the other hand, it is found that when the overestimated mode number is set to be drastically larger than the active genuine mode number, the proposed method shows degraded performance. Fig. A4 shows the case of preference parameter of 50% of the median similarity, which produced 16 candidate modes (4 times of the 4 active modes). Although the genuine modes can still be easily distinguished by visualizing the full-field mode shapes, the estimated modal frequency of the first mode (the cluster location in the frequency axis) has deviated to the neighborhood. This is because a small preference parameter has allowed too many overestimated clusters (modes) with significant noise incorporated in the APC process. Therefore, setting the preference parameter as the default median similarity has been found a very robust choice in all our experiments, which is consistent with the recommendation in Ref. [32].

3.2.3. APC "stabilization" diagram?

Since the APC method produces a robust set of the dominant active genuine modes under varying preference parameters, one may wonder whether it's feasible to produce a "stabilization" diagram for distinguishing the physical and spurious modes without the user visualization of the full-field mode shapes. Such an APC "stabilization" diagram may be constructed as varying

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preference parameters versus clusters (candidate modes). Fig. 9 shows the MAC values of the mode shapes between different preference parameters and different cluster or candidate modes. It is seen that not only the dominant active genuine modes, but also many noise, spurious modes, have high MAC values under different overestimation cases, shown as the dark red in both the genuine and spurious modes (cluster) in Fig. 9. This is because when changing the preference parameters, APC tends to only add or remove other less significant clusters to the *existing* dominant clusters, as discussed at the end of Section 2.2. Therefore, the visualization of the full-field mode shapes to distinguish the genuine and spurious modes is an essential and also user-friendly step in the present method.

3.2.4. Comparisons with existing methods (stabilization diagram of SSI, FDD & fuzzy c-means clustering)

The overestimation process promises identification of even weakly-excited modes, which is a common challenge in output-only modal analysis. In this experiment, the 4th mode (219.84 Hz) is missed by the existing PCA-CP method [12]: when observing the singular value distribution of the covariance matrix of the structural response, there is a sharp drop close to zero at the 4th singular value (Fig. 7(a)), and it is impossible to estimate active mode number by looking at the logarithm scale of the singular value distribution (Fig. 7(b)).

The stabilization diagram method combined with the SSI or PolyMAX also overestimates the modes to identify weakly-excited modes. However, the computational burden of these methods is overwhelming: (1) the parametric methods (SSI and PolyMAX) do not scale to high-dimensional data to obtain full-field, high-resolution modal parameters (mode shapes); (2) the stabilization diagram method needs to implement the system identification process for *a wide range of* order numbers (up to the dimension, which could be very high in full-field modal

analysis) to produce a stabilized set of modal frequencies (*spectral* information). In contrast, the APC method only needs to implement the system identification (overestimation) *once* to produce a set of candidate modes with associated full-field mode shapes (high-*spatial*-dimensional information). Visualizing these candidate mode shapes allows the user to easily distinguish between physical and spurious modes.

The FDD method is also a very user-friendly output-only modal identification process, allowing the user to distinguish the genuine modes by picking the dominant singular values in the spectral matrix of the covariance matrix of the structural response, as observed in Fig. A1 for this set of data. However, the computation of the spectral matrix does not scale to such a high-dimensional data either. In this experiment, computing the spectral matrix took over 7 hours. In contrast, APC computation took about 10 seconds, on the same data and same computing platform.

While APC performs efficiently for large datasets, one potential challenge arises from the fact that it uses data points as exemplars for each cluster. Thus, if the frequency resolution of the dataset is poor, the exemplars will less accurately reflect the true mode. Traditional system identification methods can work around, to some extent, low frequency resolutions by, for example, using curve fitting. For APC, low frequency resolution applications have not yet been explored. For the experimental beam vibration dataset in this study, however, the frequency resolution is relatively low at 0.8 Hz but the results by APC estimation are accurate. Although for some applications the frequency resolution could be a concern and should be the subject of future work.

Fuzzy C-means (FCM) based output-only identification method [34] was also conducted. It's known that FCM requires users to specify the cluster number beforehand. Here two cluster

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numbers were used: (1) the accurate active mode number 4 and (2) 9 over-specified mode number (for convenient comparisons with APC with preference parameter set at the default median similarity). The results are shown in Fig. A5 and Fig. A6, respectively. It is seen that except the first mode in the #2 cluster for the 4 clusters case and #5 cluster for the 9 clusters case, all other estimated mode shapes have been inaccurate, noisy, and scattering. The FCM method is seen not suitable for full-field output-only modal identification with high-dimensional data.

3.2.5. Further experimental validations – a discrete-type structure

Another experiment was conducted on a lab-scale discrete lumped-mass-type three-story structure to validate the proposed APC based output-only full-field modal identification method. Similarly, the structure was excited by a hammer impact to the top floor. The three dominant active modes are at 8.0, 23.3, and 33.3 Hz, respectively, as measured by accelerometers for reference. A 2.08 second video (1000 frames) of the vibrating structure was taken by the digital video camera the same as described in Section 3.1. Local phases are computed to produce a full-field displacement dataset. Fig. 10 shows one frame of the video along with the estimated local phase signals at three arbitrary pixels on the structure.

System identification is performed using APC and PCA-CP. The traditional FDD and PolyMax methods do not scale to this size of dataset and could not be run on a desktop computer with a 3.2 GHz processor and 64 GB RAM. By visualizing the full-field mode shapes of the *candidate* modes (clusters) in Fig. 11, the genuine modes can be easily distinguished: #1 (8.2 Hz), #3 (23.3 Hz), #5 (33.4 Hz). The associated modal frequencies of the APC modes correlate well with those estimated from accelerations, while PCA-CP identified 8.16, 23.04, and 32.12 Hz.

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3.2.6. Further experimental validations – a 2-D plate structure with closely-spaced modes

To demonstrate APC for modal identification on more complex structures and larger dataset, we established a finite element model of a 2D plate with out-of-plane excitation. The plate is $13.8 \times 10 \times 0.005$ inches with fixed corners. Mass and stiffness matrices were estimated using a 5×5 node discretization of plate elements. The corresponding 21 vibration modes lie between 7 and 76 Hz. Mass proportional damping is assigned such that the lowest frequency mode has a damping ratio of 0.01. Multiple random forces are applied to two nodes, the center node and the middle node of one side, as depicted in Fig. 12a. This excitation excites 12 modes including closely-spaced modes. State-space integration is used to solve the equation of motion, generating displacements at a sampling rate of 1000 Hz for 20 seconds. The displacements are then spatially interpolated to a 100×100 resolution to simulate full-field measurements. Finally, white Gaussian noise is added to the measurement such that the maximum signal to noise ratio is 20 dB. A schematic of the plate along with displacement signals at a few spatial locations of the simulated dataset are shown in Fig. 12.

APC is performed on the simulated plate dataset with the preference set at the median similarity, producing 74 candidate modes. Due to space limitations, a subset of these candidate modes are shown in Fig. 13. By visualizing the full-field mode shapes of the *candidate* modes (clusters) the 12 genuine modes, some of which are shown in Fig. 13, can be easily distinguished at 7.4, 23.1, 26.2, 36.1, 40.7, 44.1, 47.9, 61.1, 61.7, 65.2, 69.9, and 75.4 Hz. Comparatively, the natural frequencies of the corresponding finite element modes are 7.65, 22.97, 26.56, 35.92, 40.64, 44.00, 47.31, 61.22, 61.54, 65.00, 69.97, and 75.49 Hz. The identified 12 genuine physical mode shapes all have modal assurance criterion (MAC) values above 0.95 compared to the finite element mode shapes. Note there is a case of two very closely spaced modes at 61.22 and 61.54 Hz. These modes are both accurately identified by APC.

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3.2.7. Computational efficiency

We further investigate the computation efficiency and scalability of the APC method compared to other methods for the experimental beam and numerical plate examples. The efficiency of APC is compared to FDD [17], PolyMax [15], and PCA-CP [12] in Table 2. The datasets were processed on desktop with an Intel i7 3.2 GHz processor and 64 GB of RAM. Codes for FDD and PolyMax were modified from [35] and [36], respectively.

The results are summarized in Table 2. FDD and PolyMax were not efficient enough to processes the full-resolution plate dataset, so a spatially downsampled version of the dataset was included as well. PCA-CP, which was introduced for processing full-field datasets, along with the proposed APC-based algorithm prove scalable and efficient for processing these large datasets. On the other hand, as presented in Section 3.2.4, APC is able to identify the weak modes, which PCA-CP misses.

4. Conclusions and future work

In this proof-of-concept work we devise a new non-parametric, data-driven approach for *robust* output-only identification of high-dimensional modal parameters of basic structures, by efficiently processing and interactively exploiting the full-field measurement. Specifically, we first overestimate the system model *once* by a data-driven method affinity propagation clustering (APC), producing a pool of candidate modes associated with their modal frequencies and full-field, high-spatial-dimensional mode shapes. We then exploit and visualize the *spatial*, full-field mode shape associated with each candidate mode to distinguish the physical and spurious modes, rather than rely on the *spectral* information in the traditional stabilization diagram.

Experimental results show that the APC method provide a robust over-estimation of candidate modes, including the weak modes, without the need of a pre-determination of the cluster (mode

or model order) number. Furthermore, the full-field, high-spatial-dimensional mode shapes enable user-friendly distinction of the genuine physical and spurious modes. The advantages of the APC method in computational efficiency, scalability and performance in output-only identification of high-dimensional modal parameters were also demonstrated and compared to existing methods, including PolyMAX, FDD, PCA-CP, and FCM.

The new method was demonstrated to be versatile in successfully identifying a variety of basic structures, including a cantilever beam (continuous-type), a lumped mass discrete-type structure, and a 2D plate structure, through user-friendly straightforward visualization of the full-field candidate mode shapes produced by APC for distinction of genuine and spurious mode. The limitations of the new method exist for structures with complex geometry or shape. Since the focus of the present study is on the development and proof-of-concept, experimental study on more complex structures (e.g., the bladed wheels with peculiar mode shapes) and implementations on real-life large-scale structures (e.g., footbridges) are also needed in future work to further the devised method. In these cases, a rough picture of the mode shapes may be produced from an initial rough finite element model to augment the user for modal distinction; however, this scheme has limitations on identification of highly coalescent modesMore rigorous and more challenging, a measure of the mode shapes, of high spatial dimensional, may also be developed in the future to *quantitatively* distinguish the physical and spurious modes.

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Table 1. Modal parameters estimated from the video of the cantilever beam by the APC, PCA-CP, FDD, and PolyMax compared with the analytical solution

	Modal frequency (Hz)					Mode shape correlation with analytical (by MAC**)			
Mode	Analytical	APC	PCA-CP	FDD	PolyMax	APC	PCA-CP	FDD	PolyMax
1	7.59	7.68	7.2	7.34	7.54	0.9993	0.9979	0.9939	***
2	47.57	48	46.8	46.53	47.27	0.9684	0.9863	0.9752	***
3	133.18	132.96	133.2	134.69	133.4	0.9918	0.9257	0.9417	***
4	261.04	219.84*	--	--	220.4*	0.9501	--	--	***

Note: * Due to the sampling rate of 480 Hz, the 4th estimated modal frequency is an aliased (temporally sub-Nyquist) version of analytical modal frequency. The analytical "aliased" modal frequency is computed as 480 – 261.04 = 219.96 Hz, which is consistent with the estimated modal of APC and PolyMax.

** MAC (modal assurance criterion) values between Analytical-Video are computed using 327 points of the full-field estimated mode shapes and those of the analytical beam model.

*** PolyMAX requires a second step of performing Least Squares Frequency Domain (LSFD) for extracting the mode shapes, which can vary by different algorithms and therefore not presented here for comparisons without losing the generality of our conclusions.

-- Unable to identify the 4th mode (weak mode).

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Table 2. A comparison of computation time between modal identification algorithms.

Experiments		Cantilever beam	Discrete (lumped mass) 3-story structure	2D plate
Timesteps		600 at 480 Hz	1000 at 480 Hz	20,000 at 1000 Hz
Spatial points		2,031	9,141	10,000
Computation time (min)	PolyMax	4.51	n/a*	n/a*
	FDD	934.8	n/a*	n/a*
	APC	0.14	1.84	1.95
	PCA-CP	0.01	0.02	6.14

Note: *FDD and PolyMax could not efficiently process the full plate dataset. The required memory was insufficient on a computer with 64 GB RAM.

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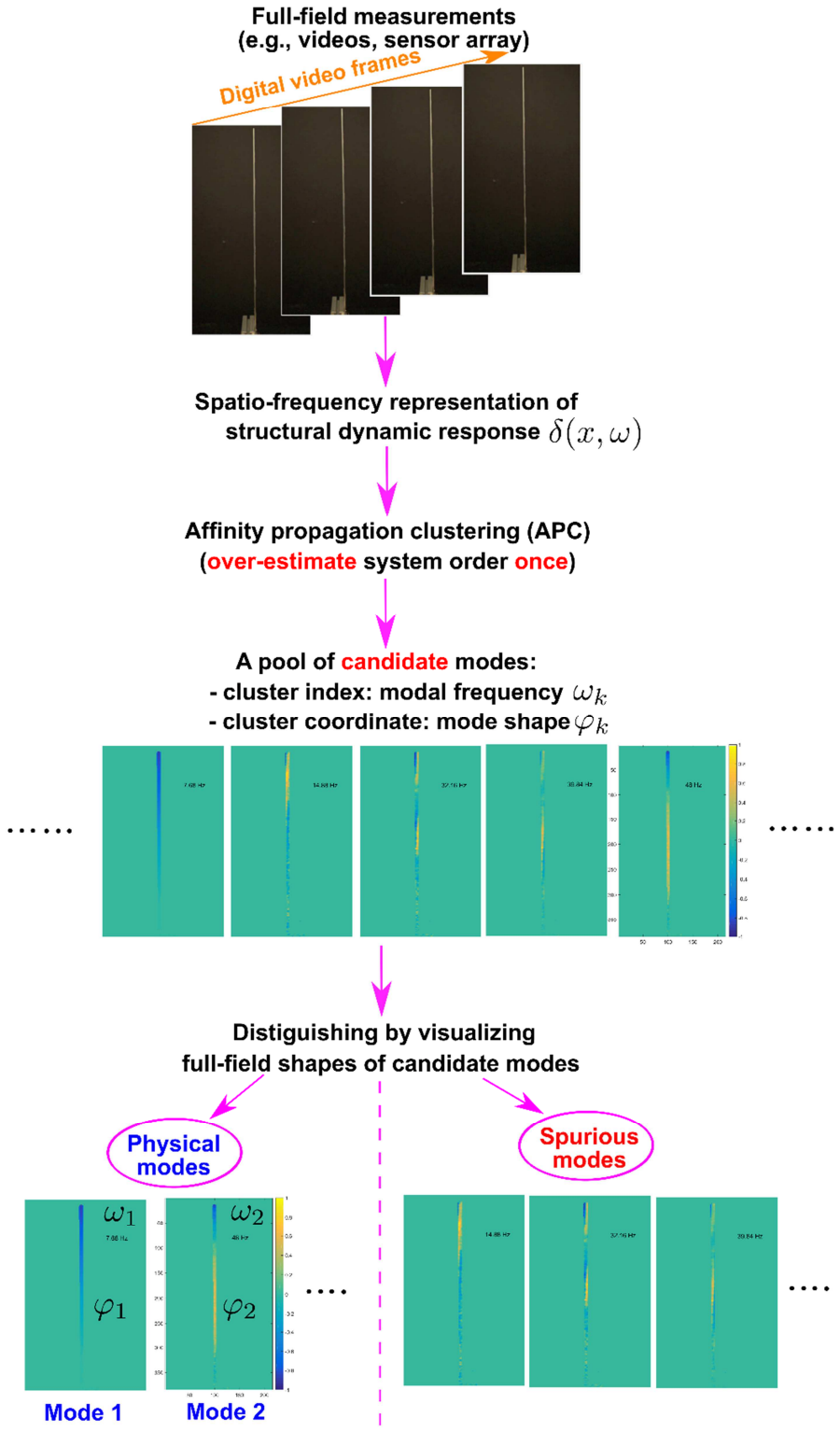


Figure 1. The flowchart of the new method for output-only identification of high-dimensional modal parameters.

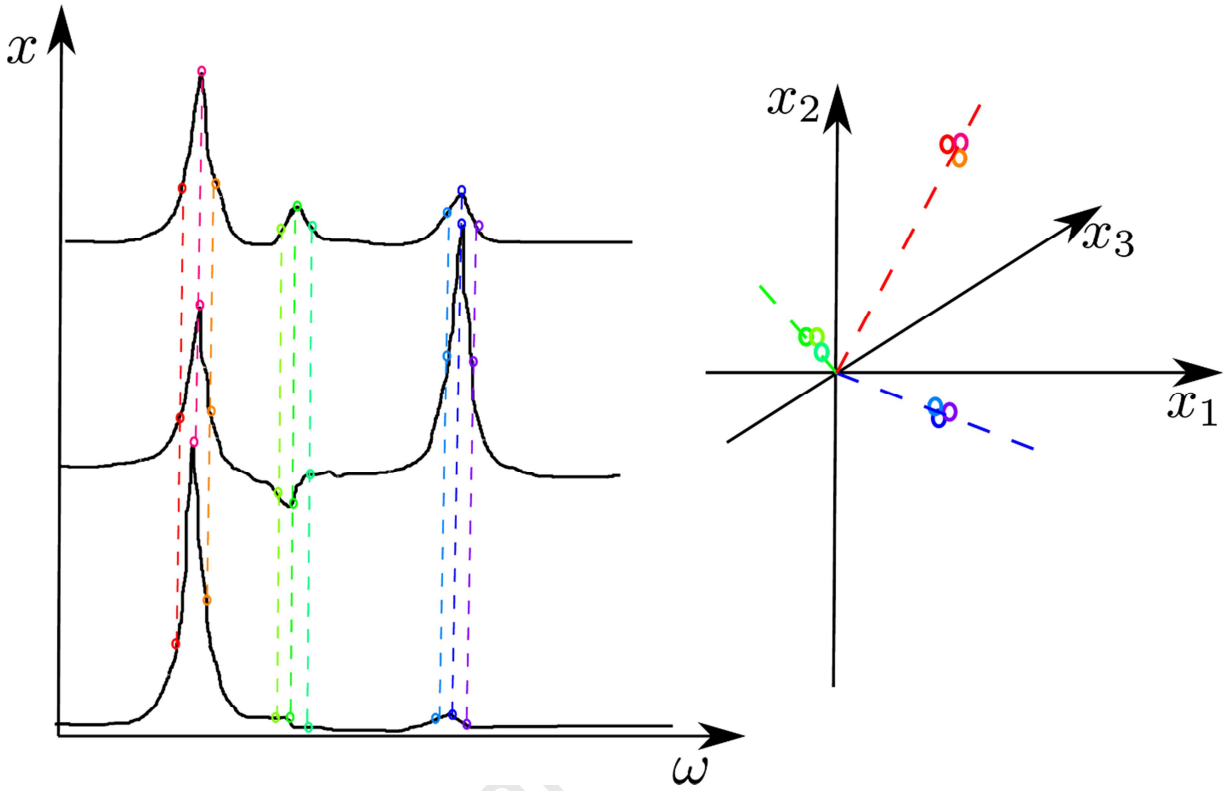


Figure 2. Left: The spatial-frequency representation of the structural response data points. **Right:** Their corresponding scattering plot (illustration is limited to 3-dimensional). The data point in this study is N-dimensional (N is high, the number of active pixels). Note that each active mode presents as a dominant *cluster* in the scatter plot: its frequency index is the modal frequency and its N-dimensional spatial representation is the mode shape vector. The weaker mode (with smaller spectral values) clusters closer to the origin filled with noise points, making it more difficult to identify.

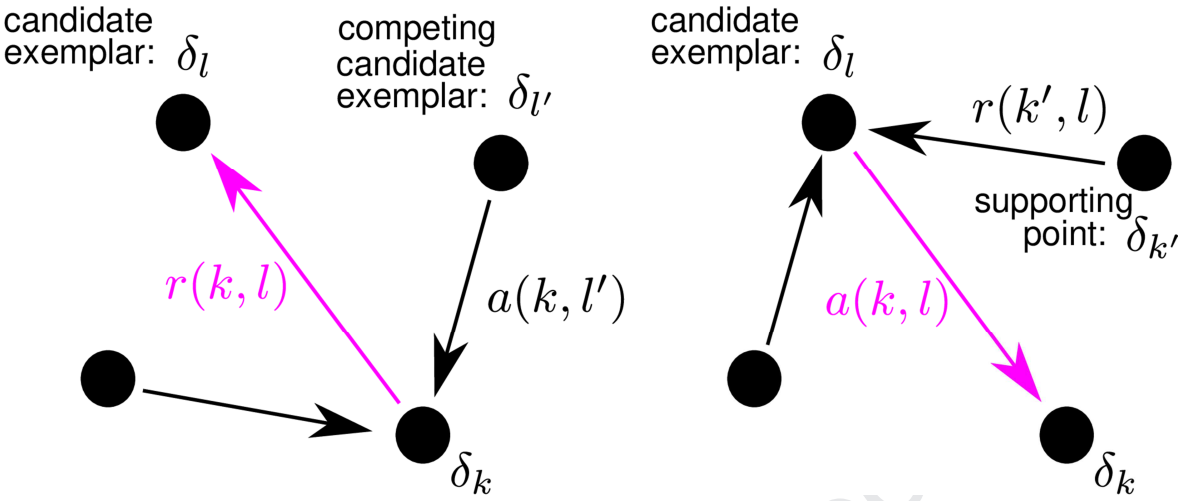


Figure 3. Left: Sending responsibilities from data points to candidate exemplars. **Right:** Sending availabilities from candidate exemplar to data points. (Adapted from [32] for illustrations)

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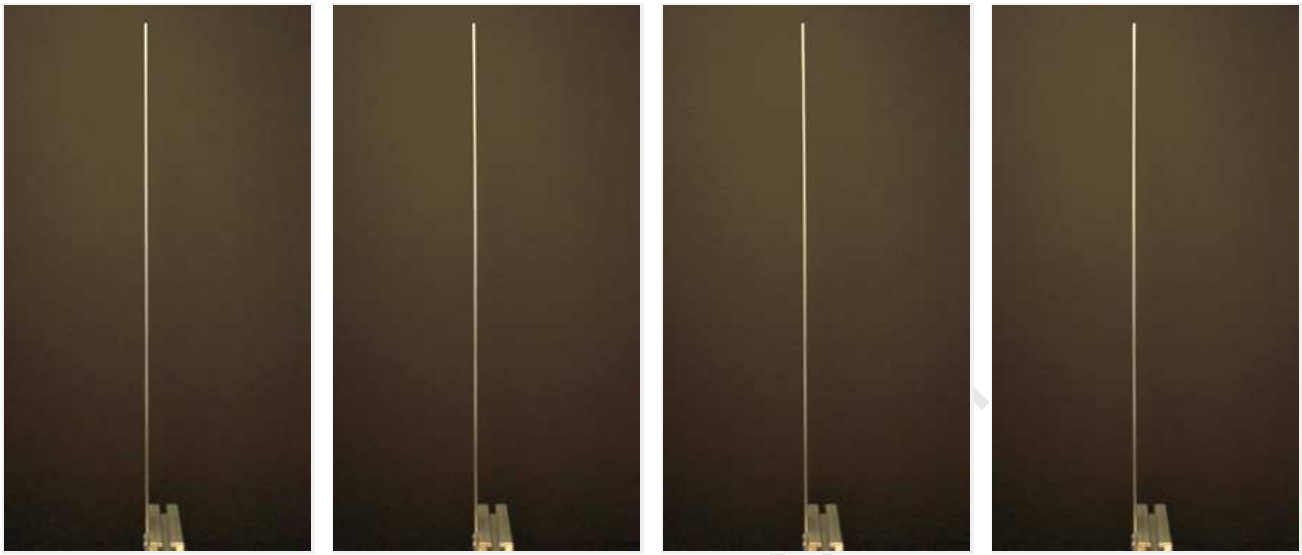


Figure 4. The full-field video measurement of a bench-scale cantilever beam excited by an impact hammer. Illustrations of a few frames of the video. The pixel (spatial) dimension is 384 (height) by 218 (width).

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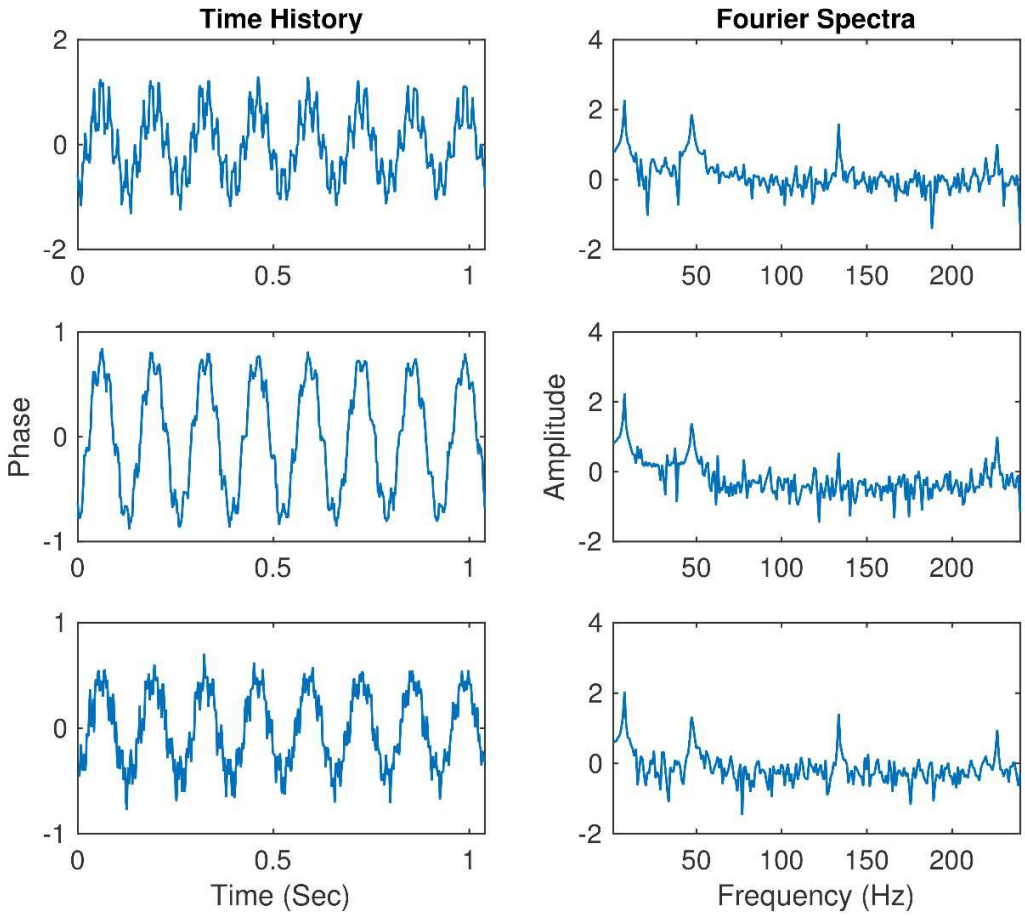


Figure 5. Illustrations of arbitrarily-selected 3 pixels: The time histories and their Fourier spectra (logarithm scale) of the extracted local phases (proportional to displacement) of 3 pixels from the video of the vibrating beam. Note that the method extracts the phase time history of **all** pixels (full-field).

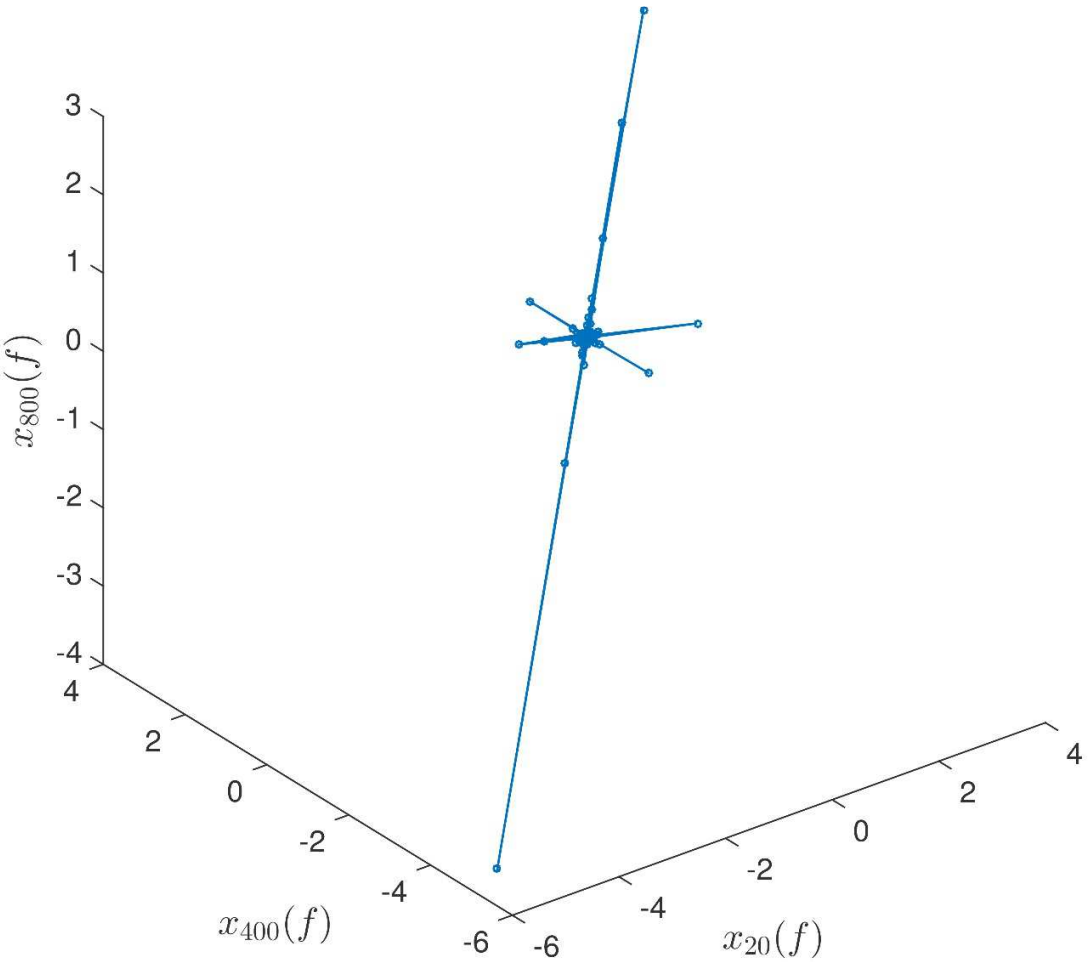


Figure 6. The scatter plots of the DCT of the phase of **arbitrarily-selected 3 pixels** (of those in Figure 3). The scatter plot is limited to 3-dimensional. The clustered vector directions are the mode shape vectors.

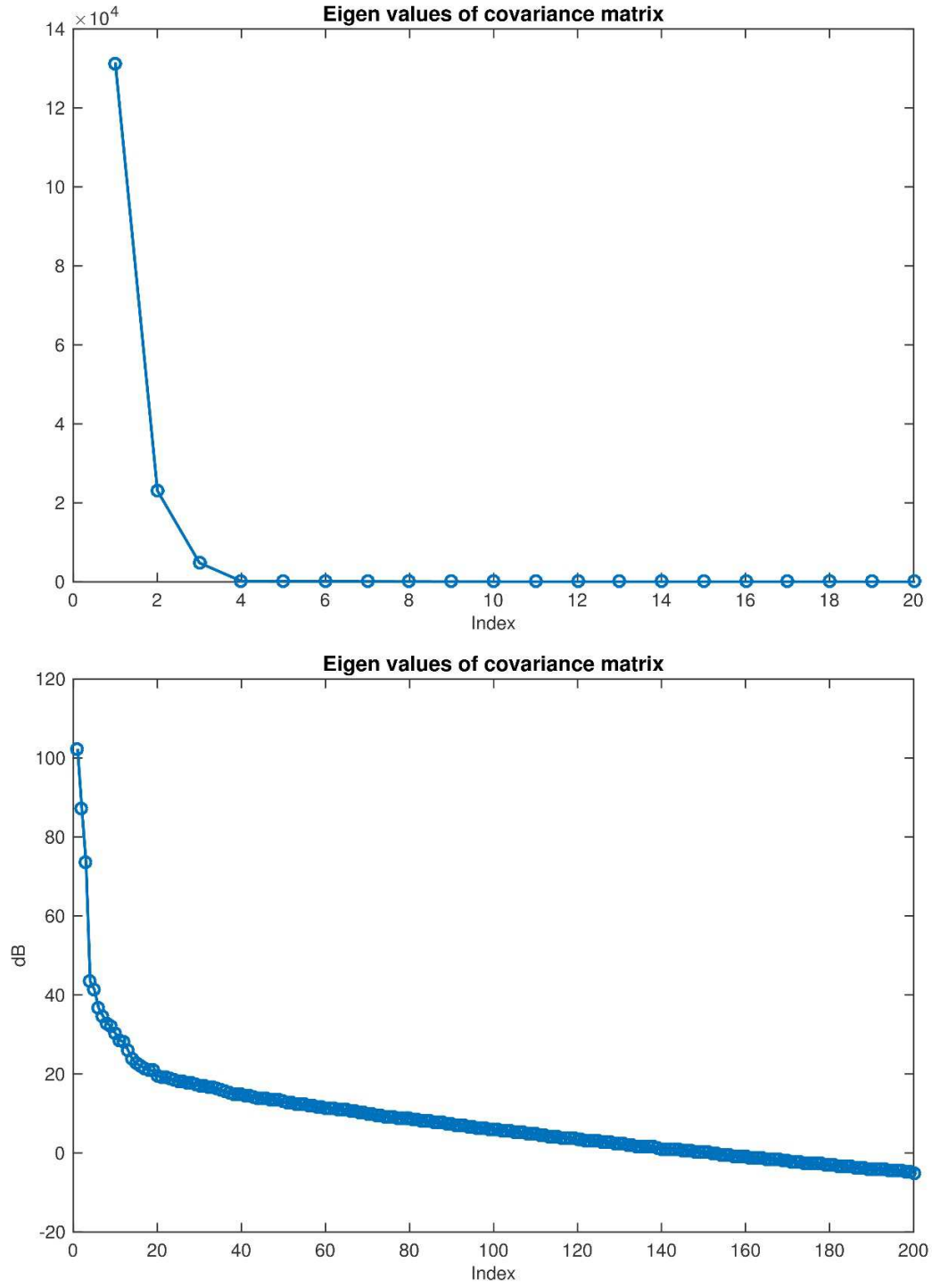
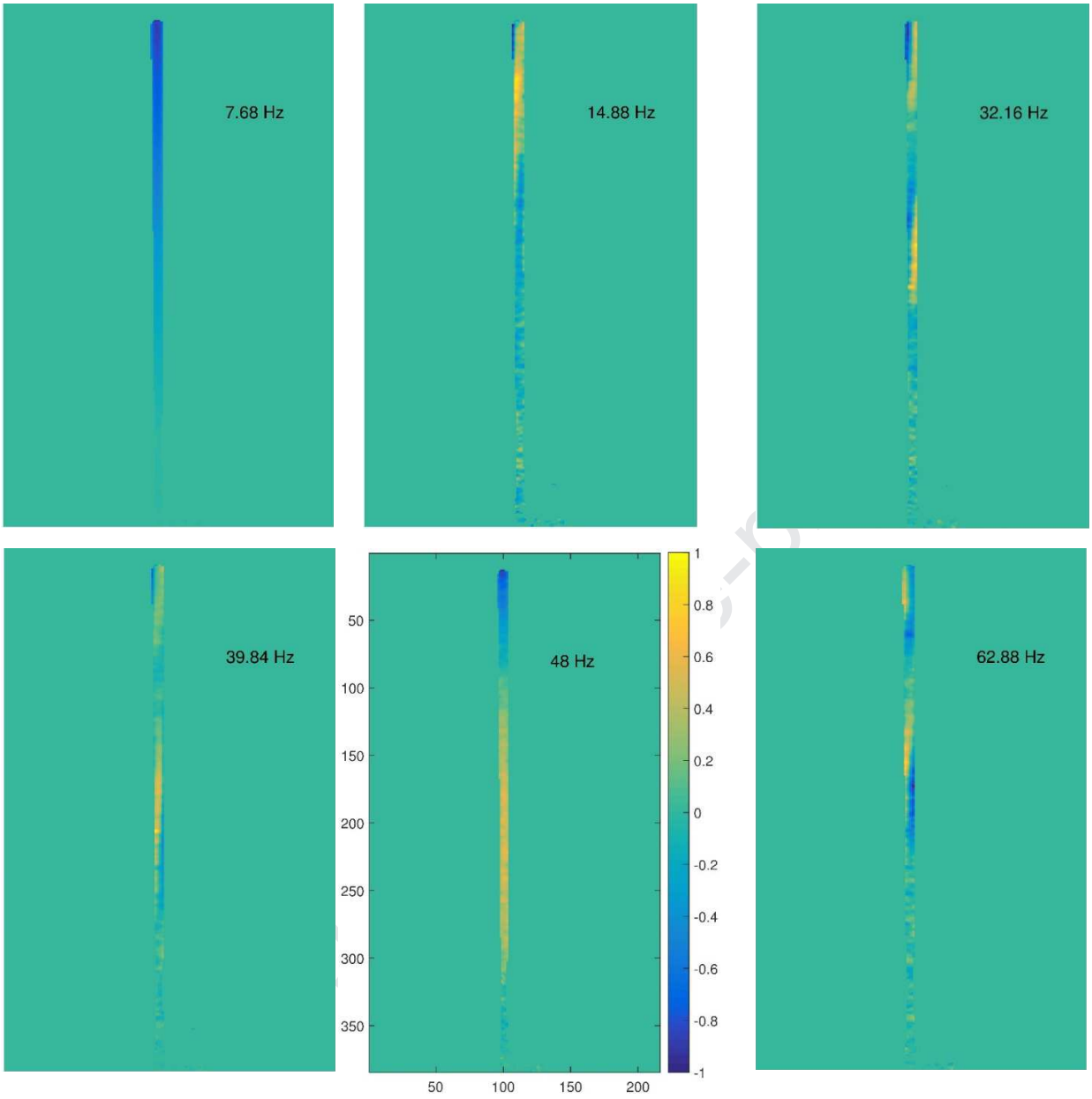


Figure 7. The eigenvalues of the structural response covariance matrix in normal and logarithm scales.


7.68 Hz
14.88 Hz
32.16 Hz
39.84 Hz
48 Hz
62.88 Hz
50
100
150
200
250
300
350
50
100
150
200
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

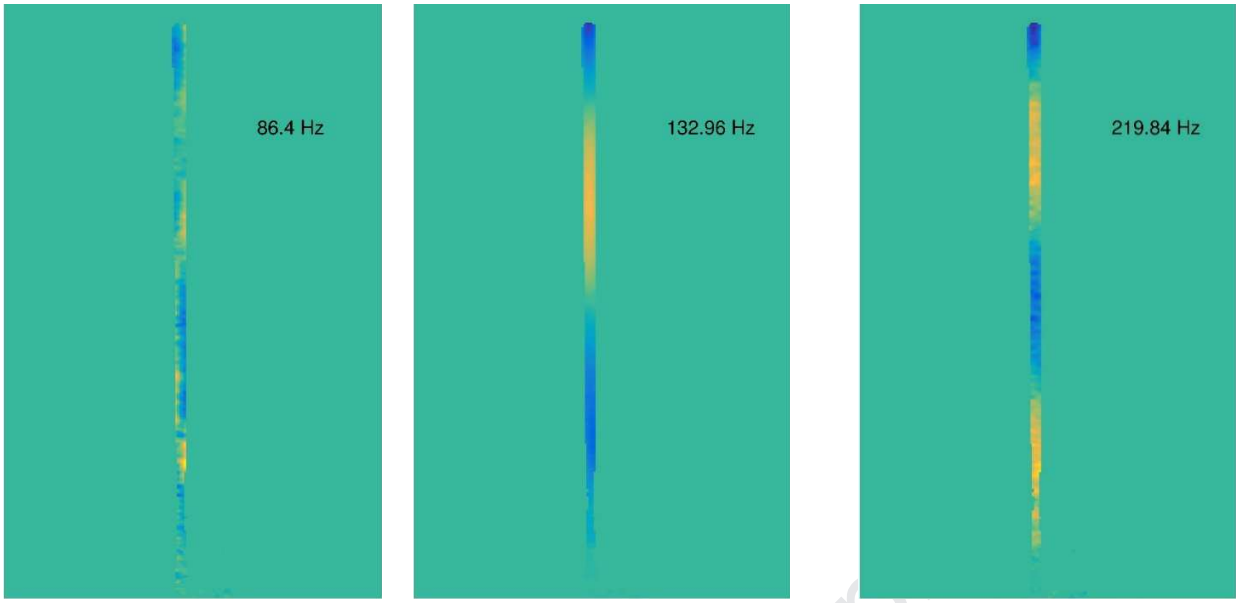


Figure 8. The APC-produced pool of candidate modes (using **default** preference setting the median similarity) identified from the video measurements of the output-only cantilever beam. Each identified candidate mode is associated with the corresponding modal frequency and full-field mode shape. The full-field displacement mode shapes in the pixel coordinate. The vertical (frame height) and horizontal (frame width) axis represent pixel locations. The colorbar (in the middle plot, same for all plots) represent normalized mode shape magnitude scales. They are of 384 x 216 pixels.

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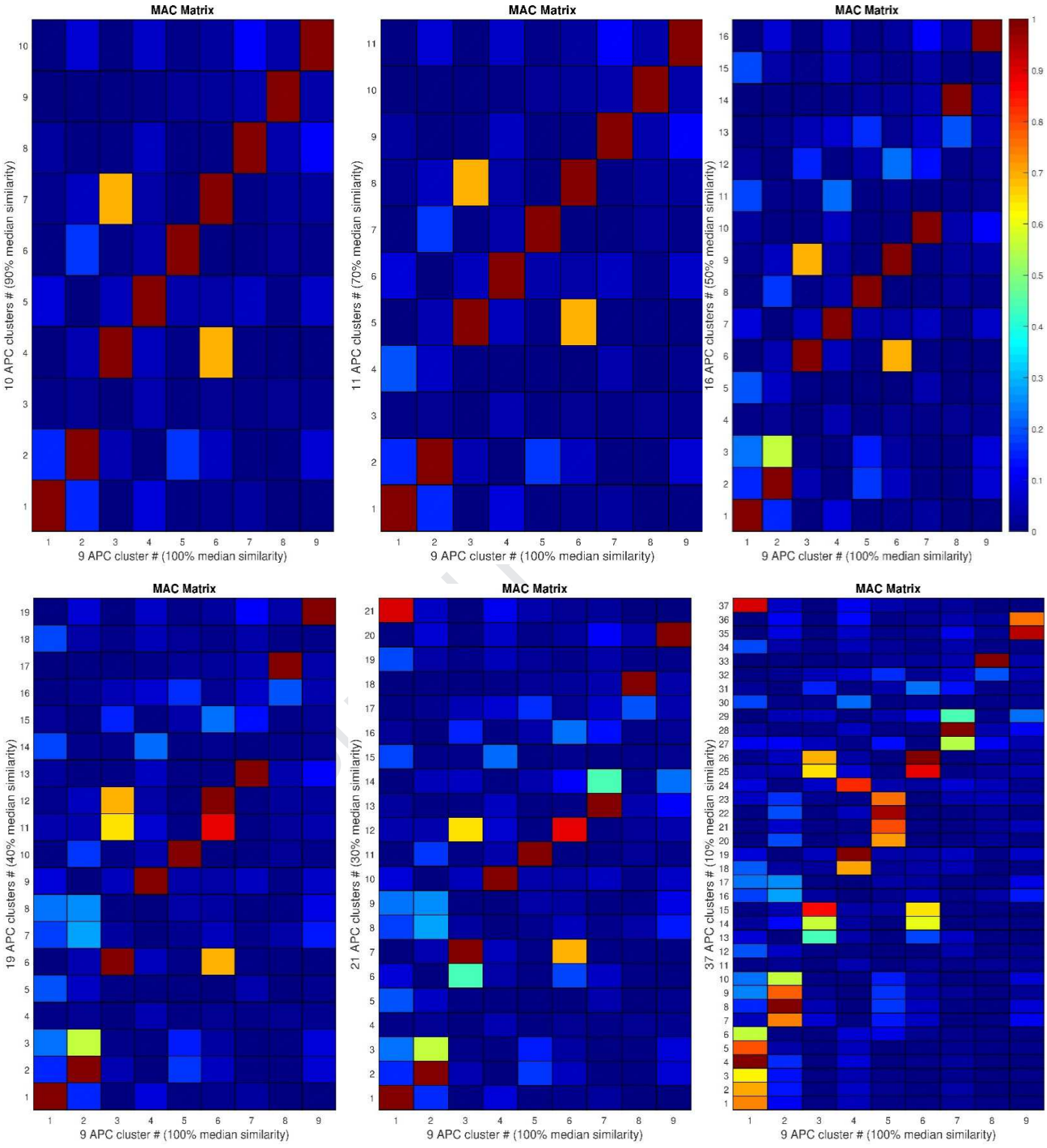


Figure 9. The correlations (MAC) among the identified mode shapes using different preference parameters in APC.

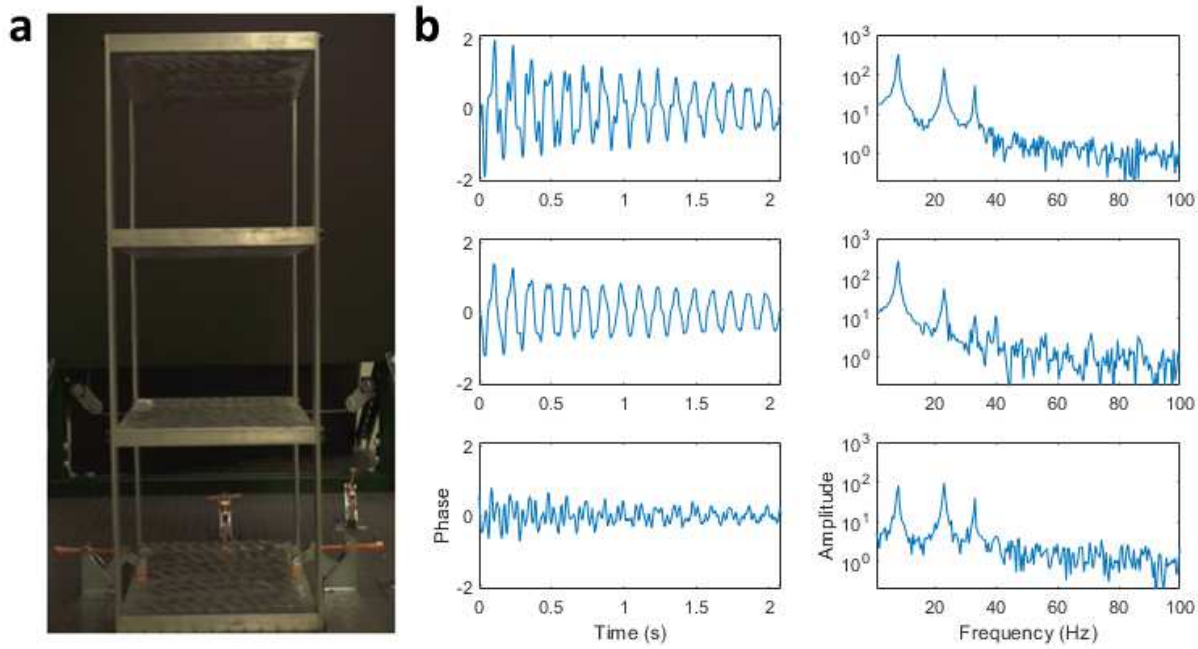


Figure 10. (a) One frame of the video measurement of the 3-story structure. The pixel (spatial) dimension is 384 (height) by 218 (width). (b) The time histories and their Fourier spectra (logarithm scale) of the extracted local phases (proportional to displacement) of 3 pixels from the video of the 3-story structure. Note that the method extracts the phase time history of **all** pixels (full-field).

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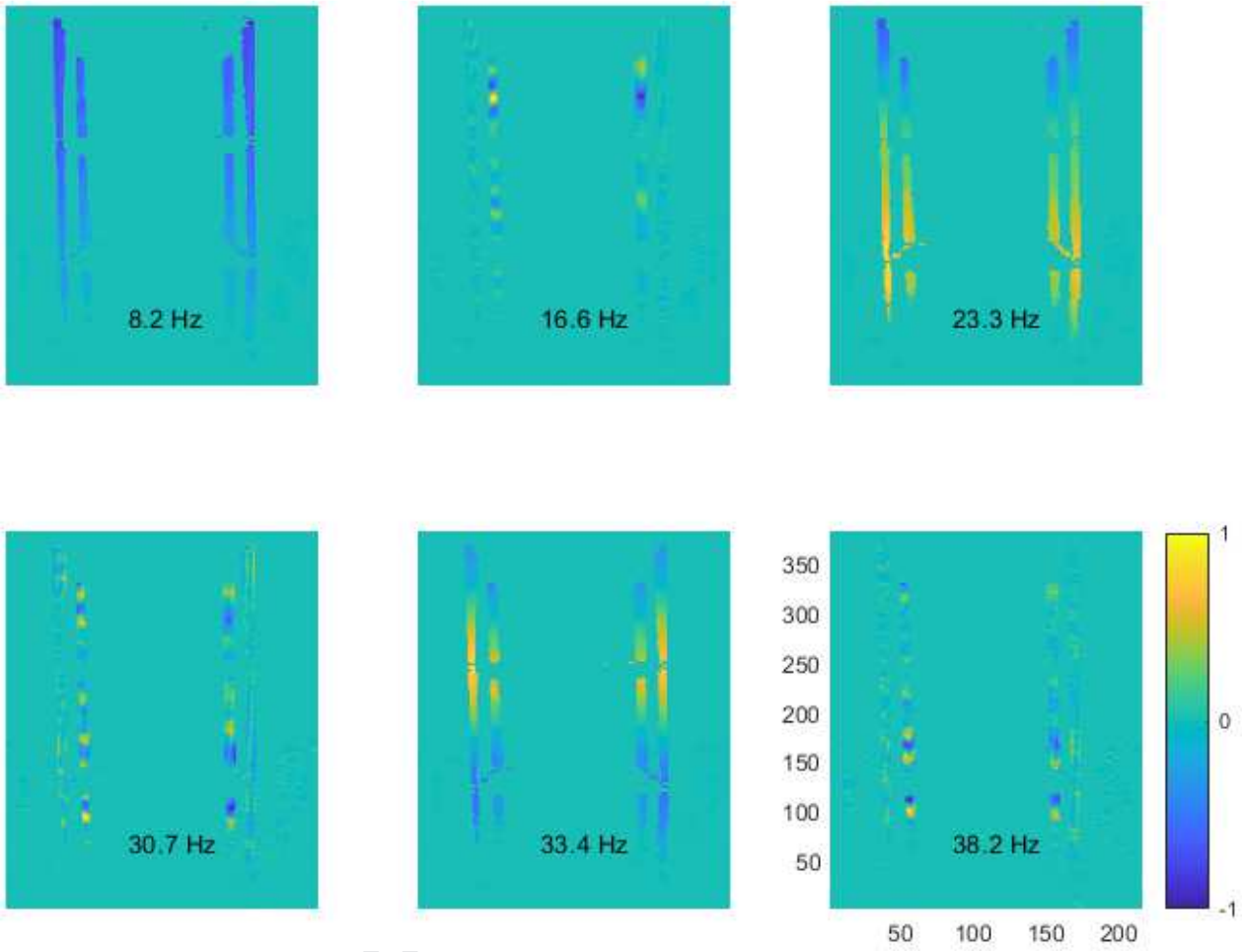


Figure 11. A few candidate modes identified by APC from 3-story structure video dataset. The vertical (frame height) and horizontal (frame width) axis represent pixel locations. The colorbar (same for all plots) represents normalized mode shape magnitude scales. They are of 384 x 216 pixels.

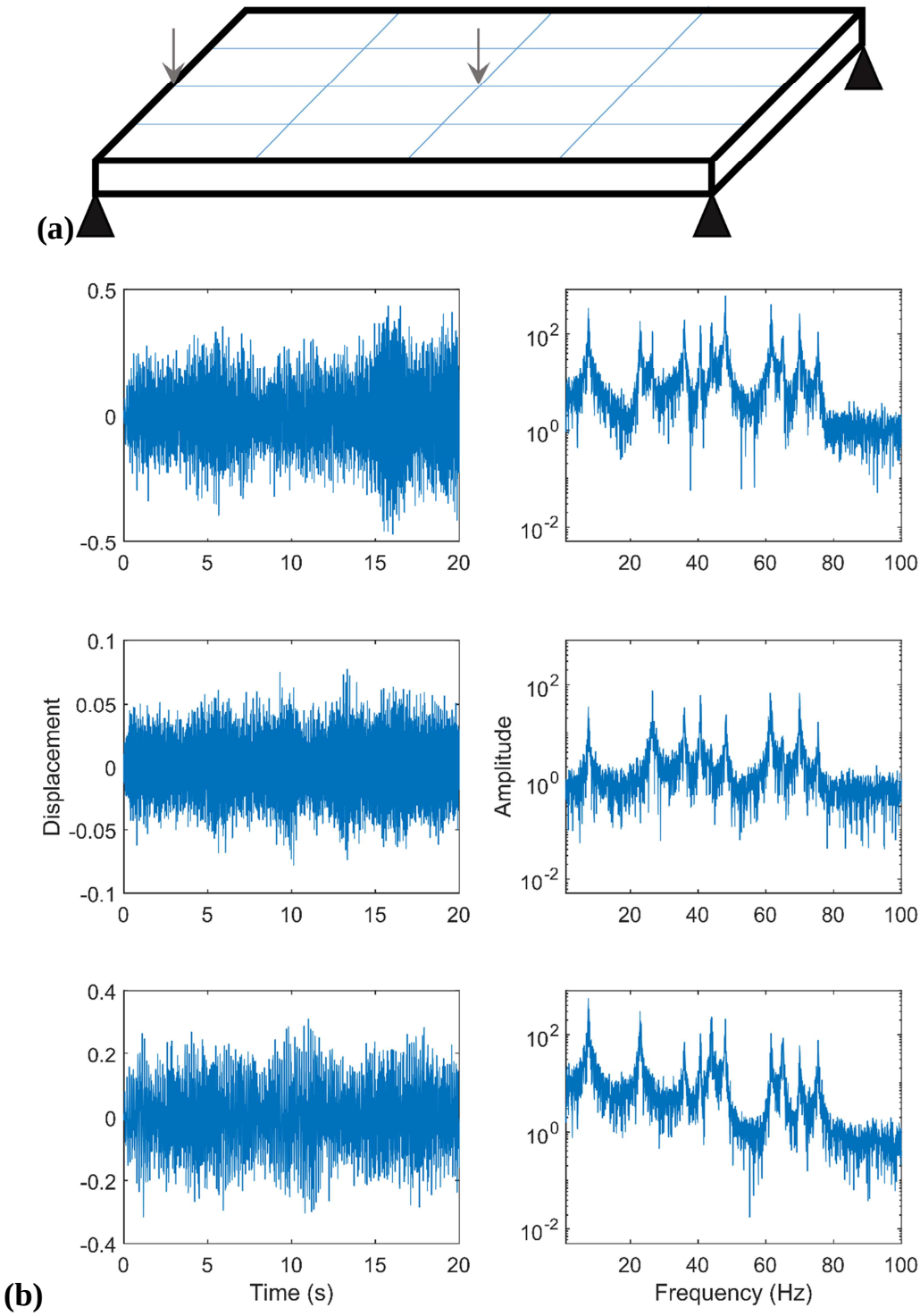


Figure 12. (a) A schematic of the numerical plate dataset, highlighting the pinned corners, 5 × 5 node discretization, and multiple random forcing (gray arrows). **(b)** A few displacement

responses of the simulated high spatial resolution (100×100) dataset and their frequency spectra.

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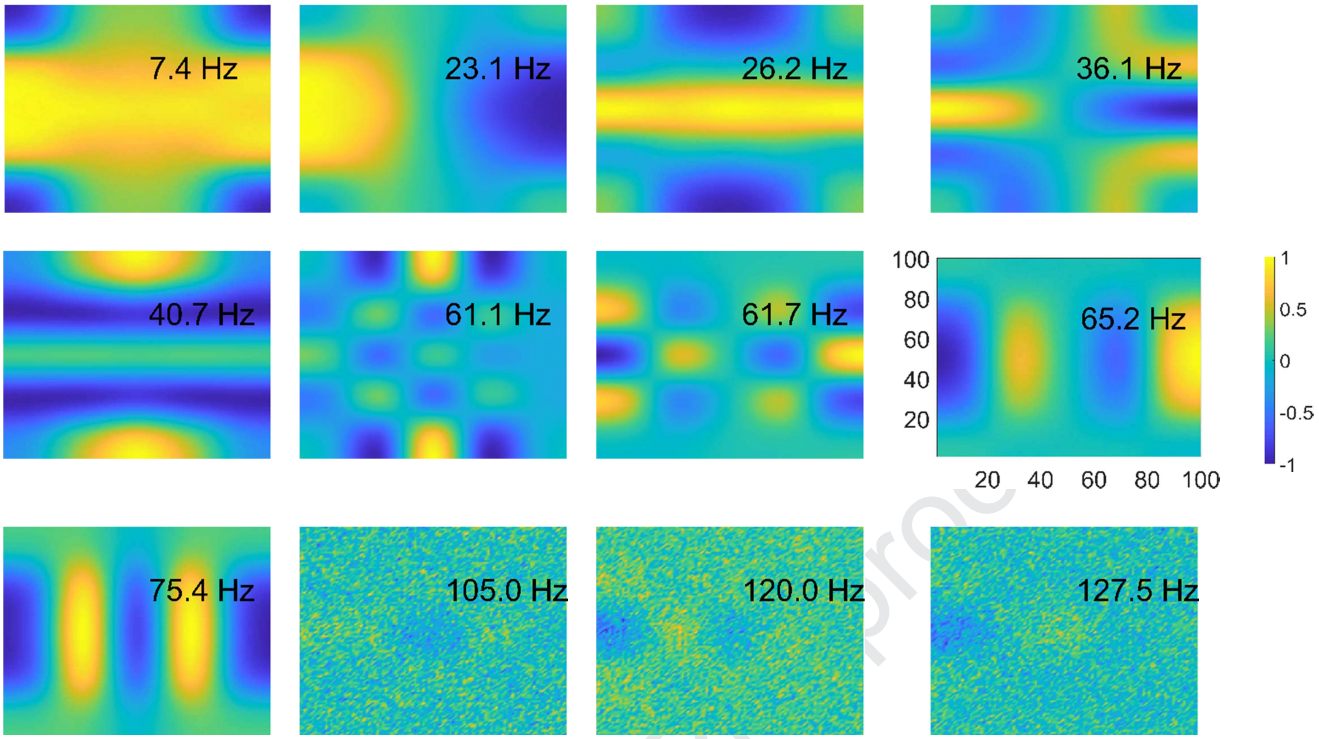


Figure 13. A few candidate modes identified by APC from the spatially high resolution 100×100 numerical plate dataset. The colorbar (same for all plots) represents normalized mode shape magnitude scales.

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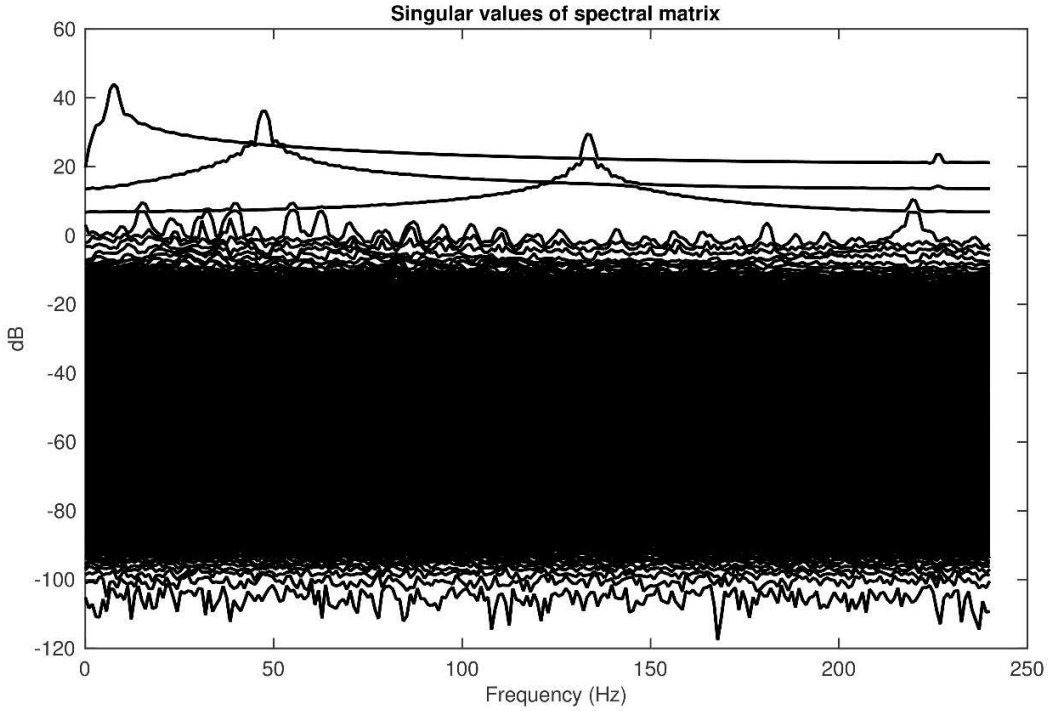
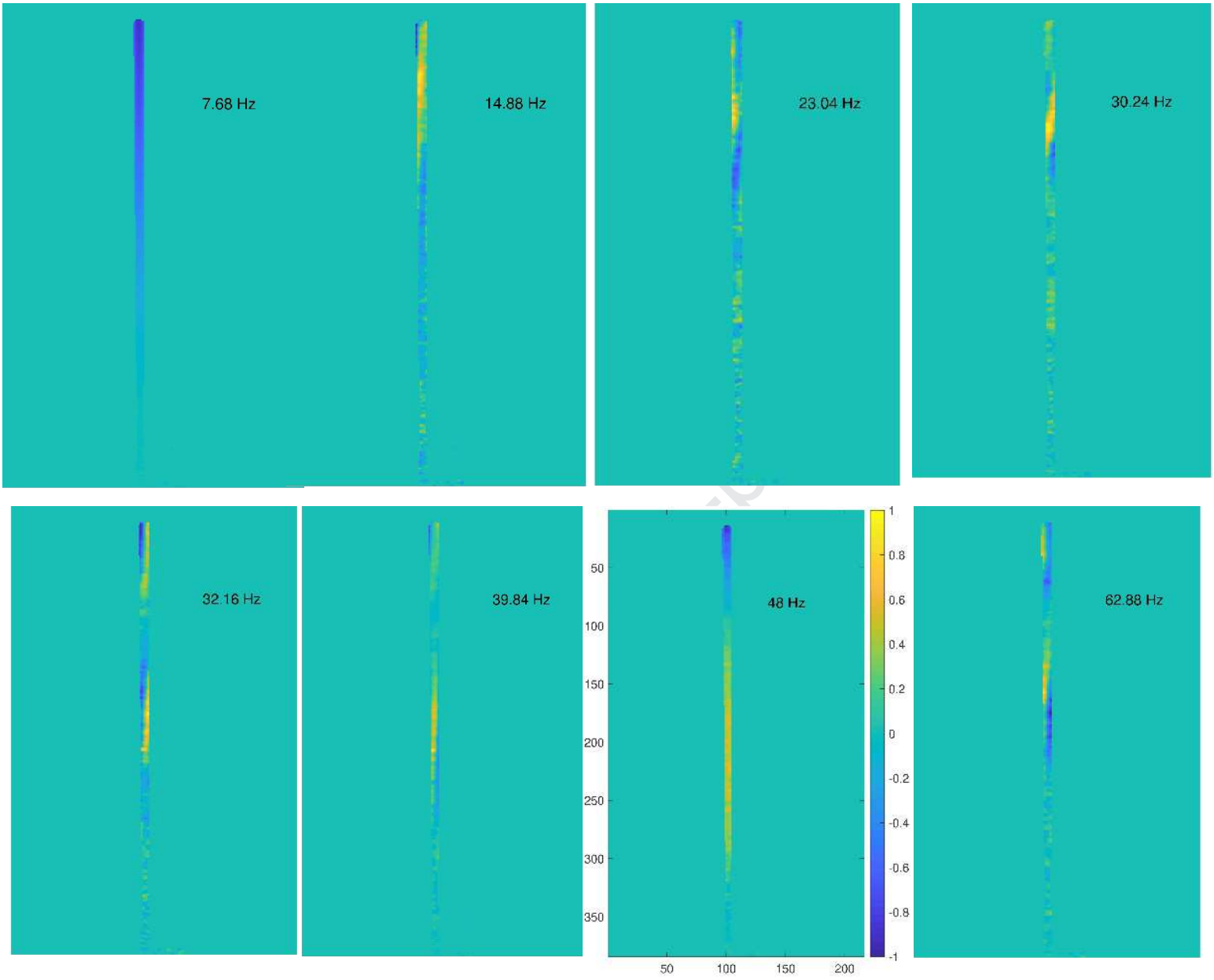


Figure A1. The singular values (logarithm scale) of the structural response spectral matrix along the frequency range. It is used by the frequency domain decomposition (FDD) method.


7.68 Hz
14.88 Hz
23.04 Hz
30.24 Hz
32.16 Hz
39.84 Hz
48 Hz
62.88 Hz
50
100
150
200
250
300
350
50
100
150
200
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

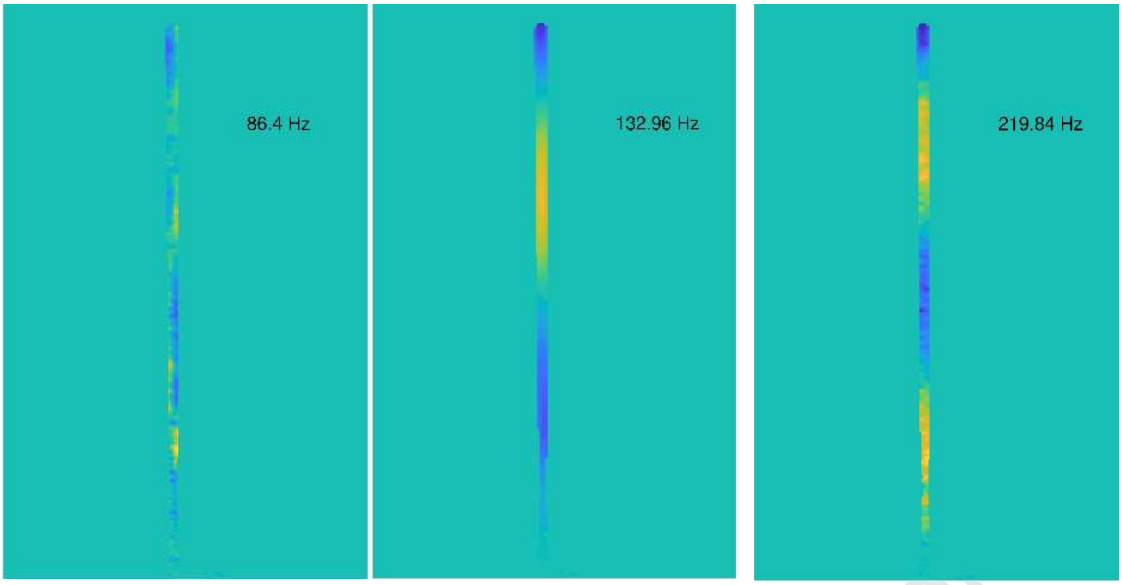


Figure A2. The APC-produced pool of candidate modes (using preference setting **70%** of the median similarity) identified from the video measurements of the output-only cantilever beam. Each identified candidate mode is associated with the corresponding modal frequency and full-field mode shape. The full-field displacement mode shapes in the pixel coordinate. The vertical (frame height) and horizontal (frame width) axis represent pixel locations. The colorbar (in the middle plot, same for all plots) represent normalized mode shape magnitude scales. They are of 384 x 216 pixels.

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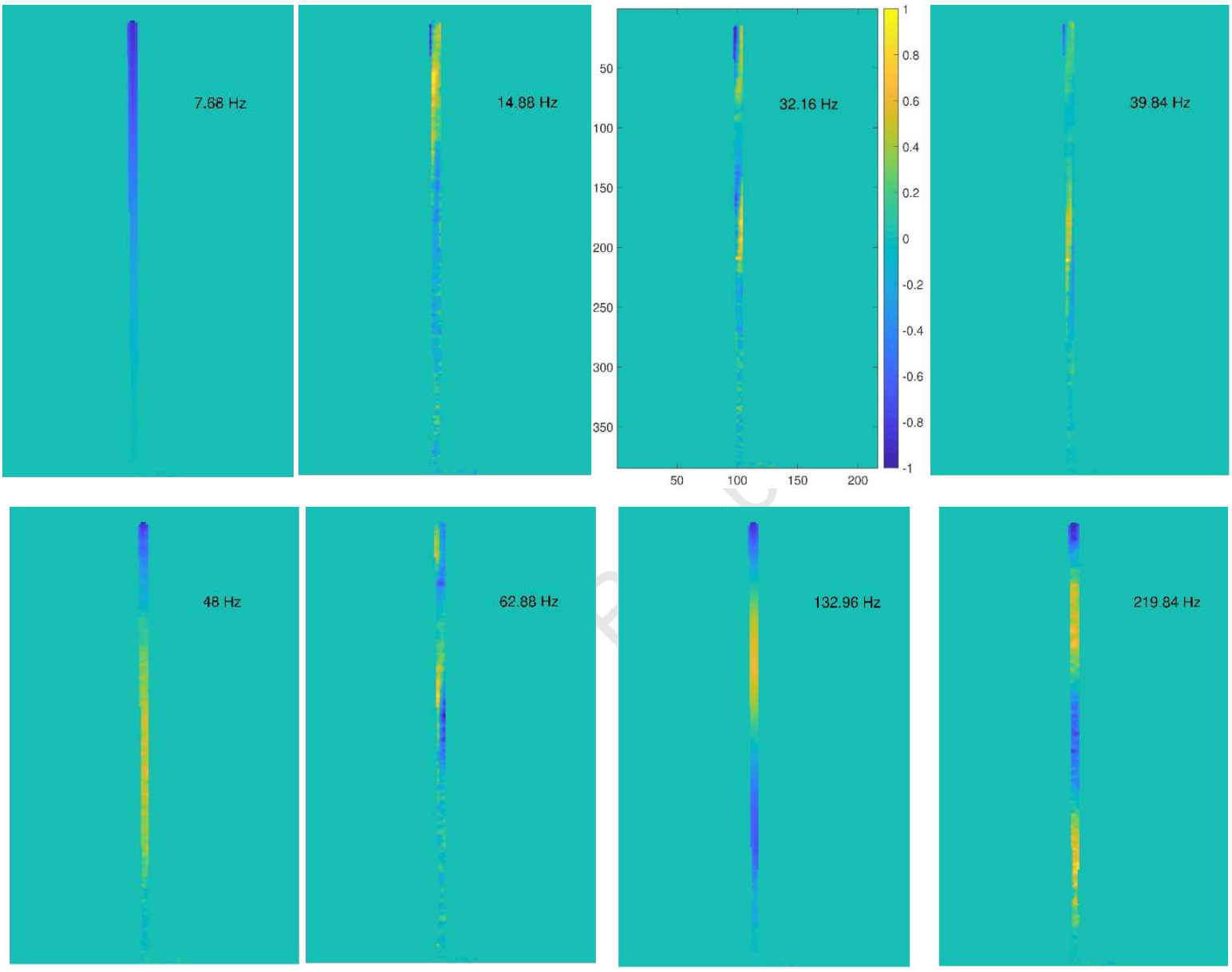
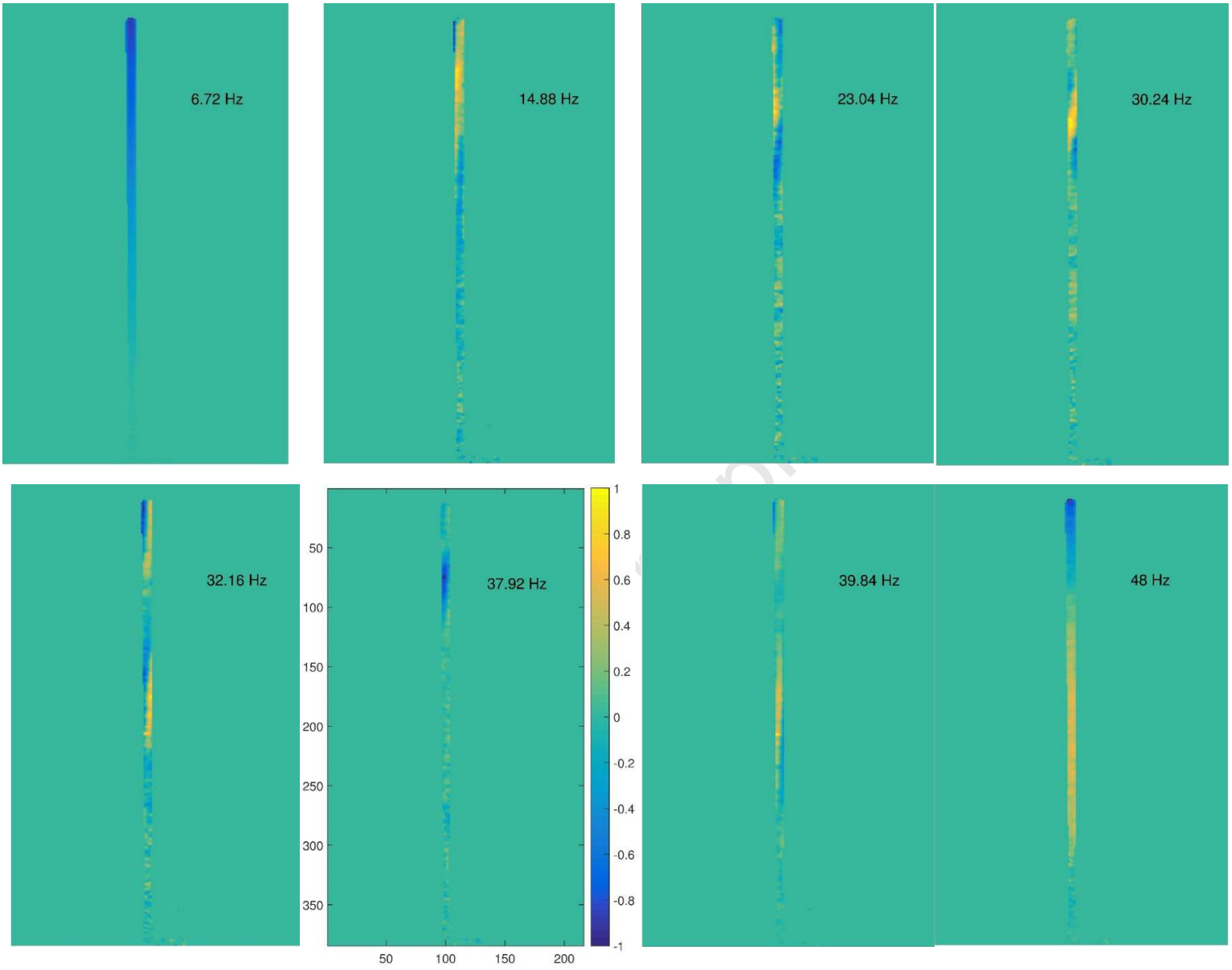


Figure A3. The APC-produced pool of candidate modes (using preference setting **150%** of the median similarity) identified from the video measurements of the output-only cantilever beam. Each identified candidate mode is associated with the corresponding modal frequency and full-field mode shape. The full-field displacement mode shapes in the pixel coordinate. The vertical (frame height) and horizontal (frame width) axis represent pixel locations. The colorbar (in the middle plot, same for all plots) represents normalized mode shape magnitude scales. They are of 384 x 216 pixels.


6.72 Hz
14.88 Hz
23.04 Hz
30.24 Hz
32.16 Hz
37.92 Hz
39.84 Hz
48 Hz
50
100
150
200
250
300
350
50
100
150
200
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

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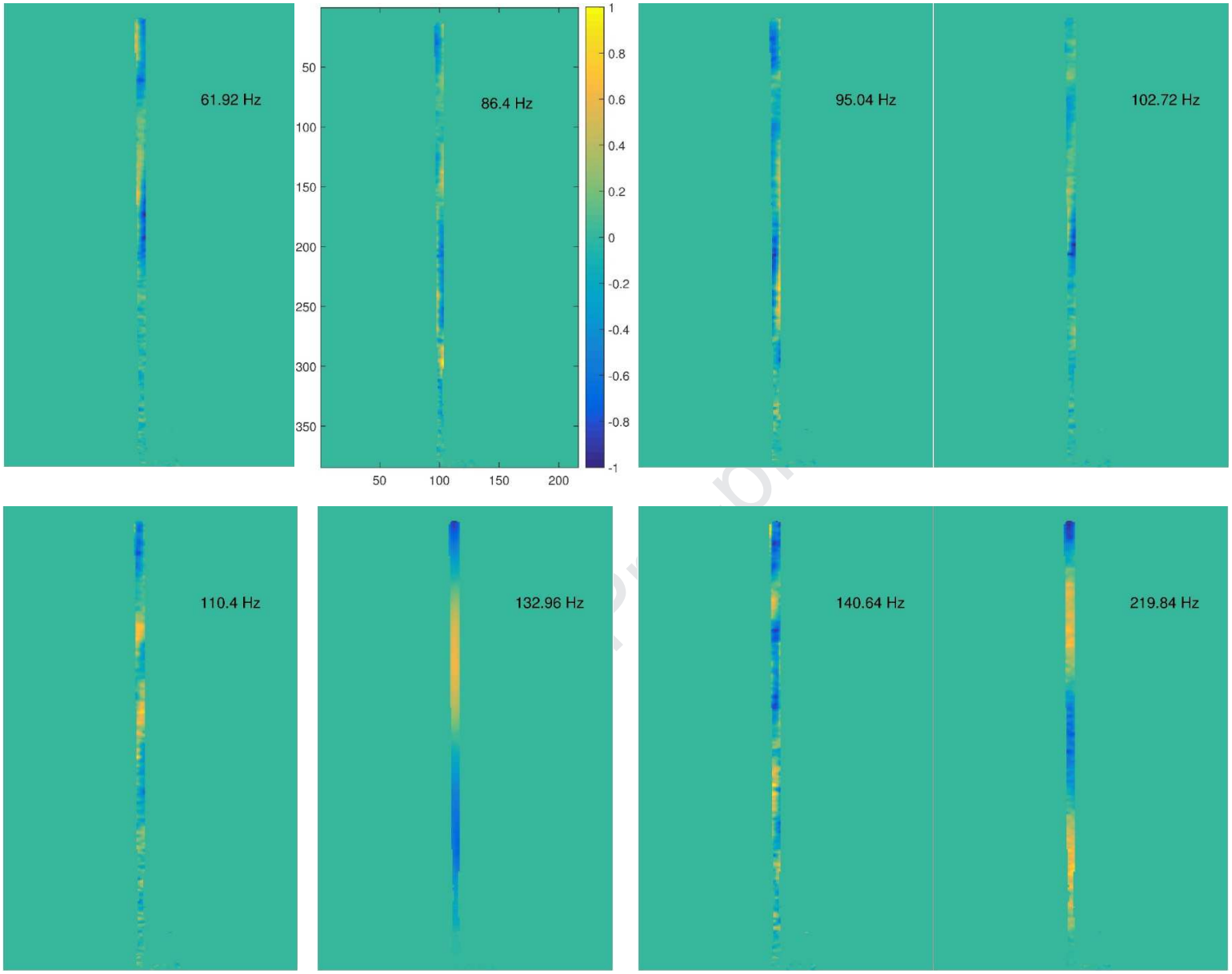


Figure A4. The APC-produced pool of candidate modes (using preference setting **50%** of the median similarity) identified from the video measurements of the output-only cantilever beam. Each identified candidate mode is associated with the corresponding modal frequency and full-field mode shape. The full-field displacement mode shapes in the pixel coordinate. The vertical (frame height) and horizontal (frame width) axis represent pixel locations. The colorbar (in the middle plot, same for all plots) represents normalized mode shape magnitude scales. They are of 384 x 216 pixels.

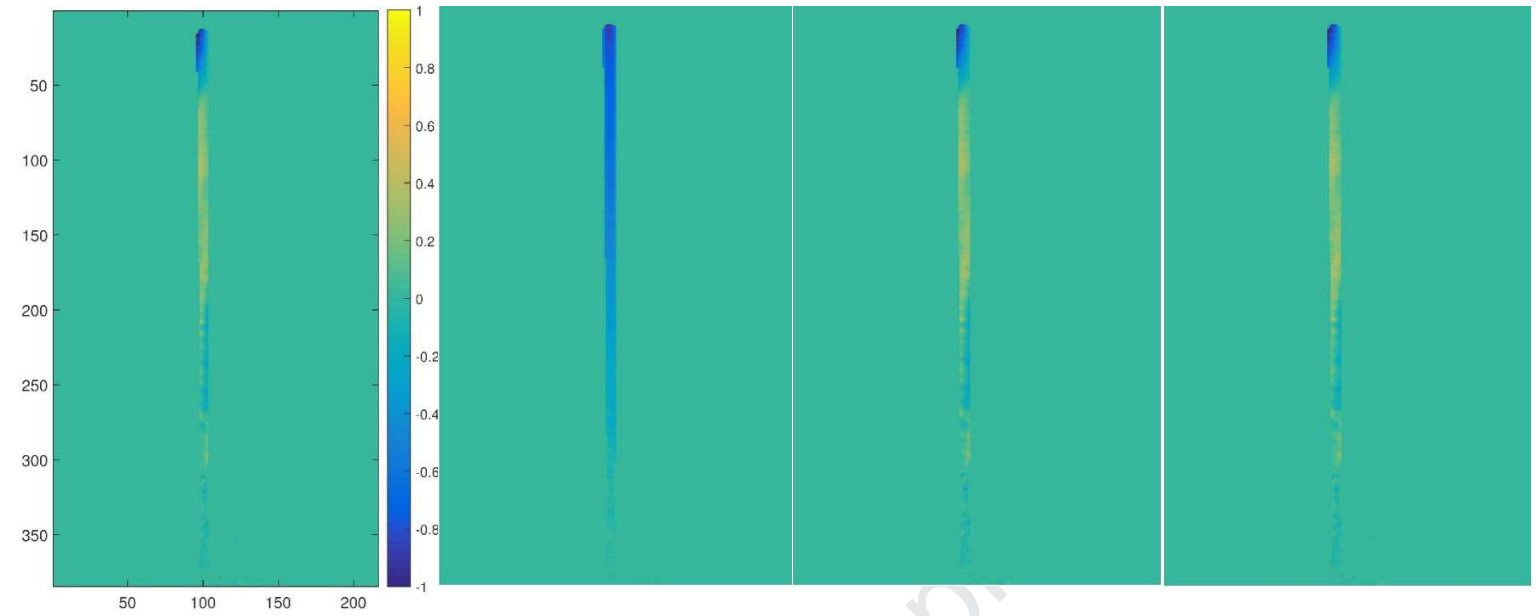
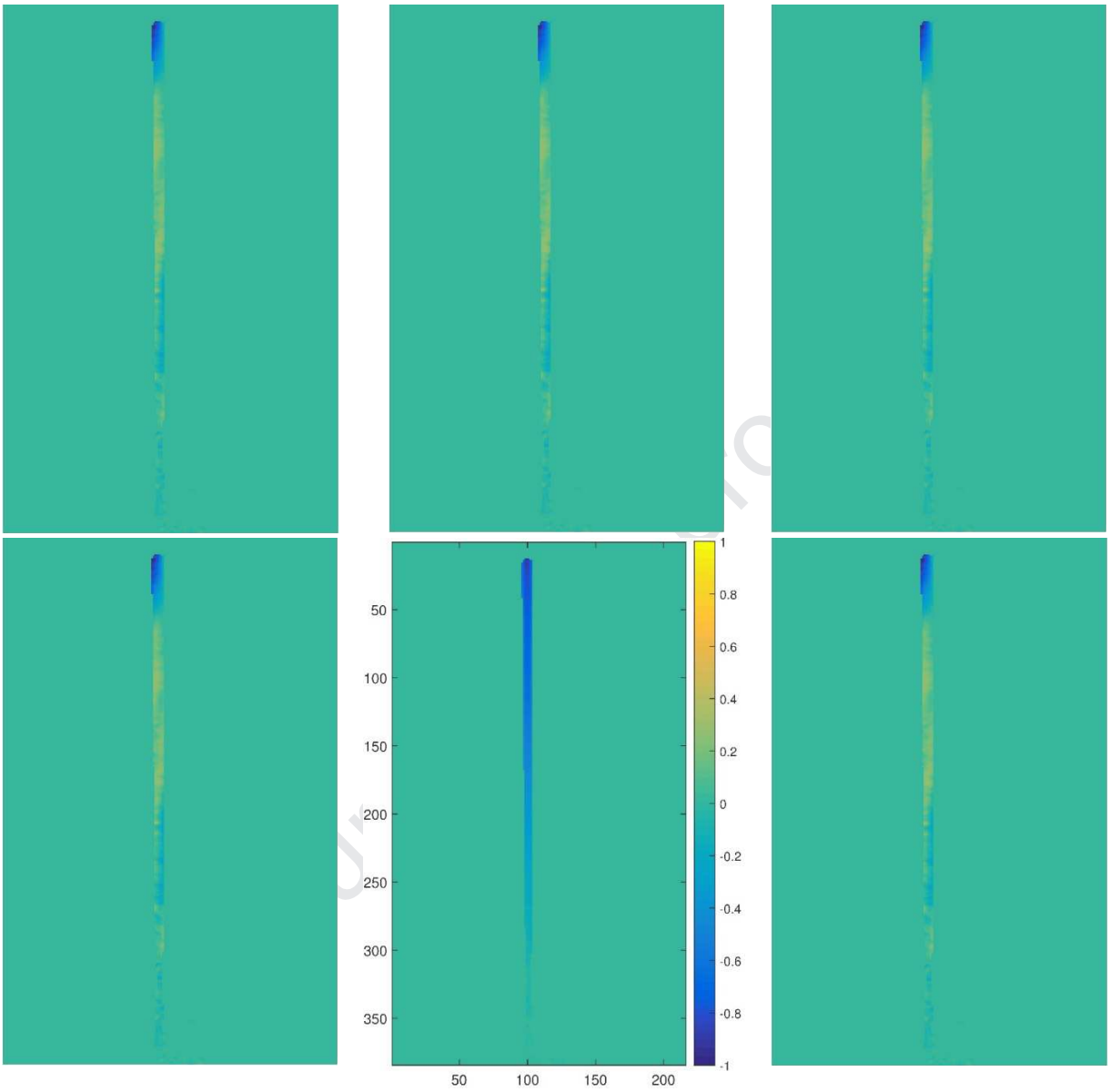


Figure A5. The full-field mode shapes produced by the **fuzzy-c-means (FCM) with a pre-specified cluster number of 4.** The full-field displacement mode shapes in the pixel coordinates. The vertical (frame height) and horizontal (frame width) axis represent pixel locations. The colorbar (in the middle plot, same for all plots) represents normalized mode shape magnitude scales. They are of 384 x 216 pixels.

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50
100
150
200
250
300
350
50
100
150
200
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1

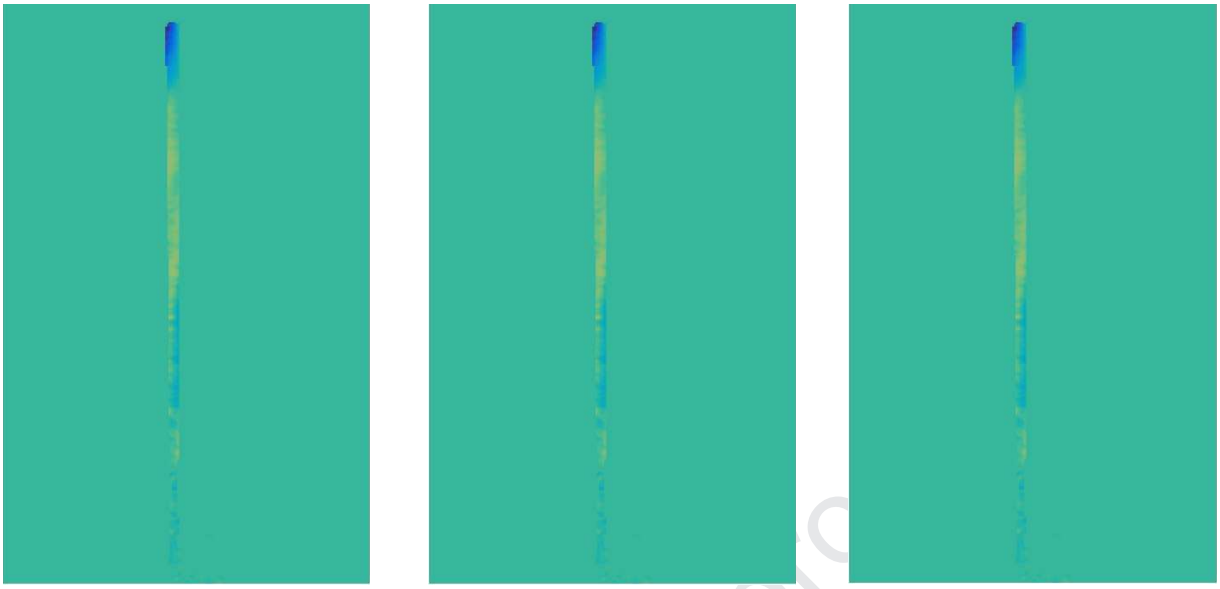


Figure A6. The full-field mode shapes produced by the **fuzzy-c-means (FCM) with a pre-specified cluster number of 9 (over-estimation).** The full-field displacement mode shapes in the pixel coordinates. The vertical (frame height) and horizontal (frame width) axis represent pixel locations. The colorbar (in the middle plot, same for all plots) represents normalized mode shape magnitude scales. They are of 384 x 216 pixels.

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Highlights:

1. Develops a new method for output-only identification of high-spatial-dimensional (very dense spatial measurement points) modal parameters.

2. Efficient processing of high-dimensional data using a data-driven method affinity propagation clustering.

3. Robust and able to extract weakly-excited modes through overestimation of the model order only once and interactive visualization of the full-field spatial information of structures.

4. Validated by laboratory experiments and extensive comparisons with existing methods (SSI, FDD, and other clustering based method).

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Author contributions: Y.Y. and C.D. jointly developed the idea. Y.Y. and C.D. performed mathematical formulations and analyses. Y.Y. and C.D. contributed to the manuscript.

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We declare no conflict of interest.

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